CHE 201 Spring 2020 Quiz 1

Thanh Nguyen

TOTAL POINTS

100 / 100

QUESTION 1

1 Question 1 **60 / 60**

- ✓ **- 0 pts** Correct
- **- 10 pts** Patm is wrong
- **- 20 pts** formulas are wrong
- **- 5 pts** unit is wrong
- **- 5 pts** mistake with the formula
- **- 60 pts** not done
- **- 10 pts** calculation error

QUESTION 2

2 Question 2 **40 / 40**

- ✓ **- 0 pts** Correct
- **- 5 pts** units are missing
- **- 5 pts** system boundary not labeled
- **- 40 pts** not done
- **- 5 pts** P and V values not marked
- **- 10 pts** state 2 is wrong

CHE 201 INTRO TO THERMODYNAMICS **January 29th, 2020**

NAME: Thanh Nguyen **UIN:** 65005534 8

QUIZ 1: 2 problems (10 pts)

Closed book/Closed notes: Show all your work for full credit

1. (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{atm}=101.325(1-0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in kPa and z is altitude in km with z=0 at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver (z=1610m). The fluid has a density of 850kg/m³ and the monometer column height is 200cm. Determine the absolute and gage pressure of gas within the tank.

2. (4 pts) A gas contained within a piston-cylinder assembly undergoes two processes in series:

Process 1-2: Constant-pressure compression at 1 bar from $V_1 = 2.5$ m³ to $V_2 = 1$ m³

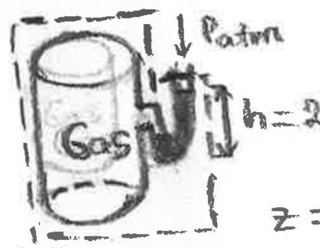
Process 2-3: Constant volume compression from 1 bar to 2 bar

a. Draw a schematic of the system and label the system boundary.
b. Sketch the processes on a p-V diagram; label States 1, 2 and 3 and the pressure and volume values at each numbered state.

① Known: $P_{atm} = 101.325(1-0.02256z)^{5.256}$, z is the altitude in km; a monometer is used to measure the pressure of a gas in a tank in Denver (z = 1610 m). Fluid density is 850 kg/m³

Find: the absolute and gage pressure of the gas in the tank

Sketch



Engineering Model

a. $g = 9.8$ m/s²

b. The system is closed

Analysis

$h = 200$ cm $= 2$ m ; $z = 1610$ m $= 1.61$ km

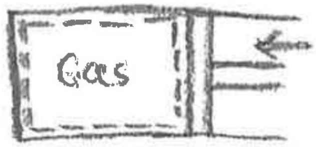
$P_{atm} = 101.325[1-0.02256(1.61)]^{5.256} = 83.4$ kPa

$P_{absolute} = P_{atm} + \rho g h = (83.4 kPa) + (850 kg/m^3)(9.8 m/s^2)(2m)$

$= (83.4 kPa) + 16660 Pa = 83.4 kPa + 16.66 kPa = 100.06 kPa$

→ $P_{gage} = P_{absolute} - P_{atm} = 100.06 kPa - 83.4 kPa = 16.66 kPa$

② a. Sketch



b. Diagram